

AITUTAKI, Cook Islands

WMO: 918310

Lat: 18.833S

Lon: 159.767W

Elev: 16

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	68.1	69.4	54.4	62.9	70.8	56.1	67.1	71.5	25.5	73.3	23.5	73.9	8.5	140	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	86.5	79.4	85.7	79.0	84.9	78.5	80.9	84.9	80.1	84.1	79.6	83.4	9.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	154.1	83.8	79.0	150.4	83.1	78.3	147.2	82.5	44.5	85.1	43.8	84.5	43.1	83.8	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	20.7	18.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	79.5	80.8	80.8	79.7	77.7	76.2	75.6	74.5	75.5	76.6	78.6	79.2	
	DBStd	2.96	2.10	2.14	1.84	1.99	2.09	2.21	2.39	2.49	2.15	1.97	1.95	1.95	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	10173	916	862	954	891	859	787	792	761	764	824	858	905	
	CDD65	4698	451	442	489	441	394	337	327	296	314	359	408	440	
	CDH74	33273	3903	4321	4729	3862	2591	1664	1456	1014	1266	1871	3014	3582	
	CDH80	4583	631	947	993	606	197	71	47	34	56	108	375	518	
Wind	WSAvg	9.0	8.2	8.8	6.7	8.4	8.9	9.0	10.7	9.3	9.4	10.1	9.4	8.7	
Precipitation	PrecAvg	77.3	10.2	9.3	8.4	9.0	7.3	3.5	3.1	3.7	3.3	4.6	6.0	8.9	
	PrecMax	108.9	20.0	19.6	19.3	23.8	19.5	10.6	8.4	11.3	17.0	13.1	16.1	21.7	
	PrecMin	42.1	1.2	1.3	1.1	2.1	0.6	0.5	0.4	0.2	0.3	0.7	0.7	0.2	
	PrecStd	15.6	4.5	4.4	4.4	5.2	4.8	2.8	2.2	2.9	3.4	2.8	3.9	5.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.8	87.6	87.7	86.5	84.4	83.2	82.7	82.7	83.0	83.8	85.9	86.3	
		MCWB	79.8	79.5	80.6	79.4	78.3	77.8	77.3	76.8	76.1	76.0	76.6	79.6	
	2%	DB	85.4	86.4	86.3	85.4	83.1	81.7	81.4	81.0	81.6	82.4	84.5	84.7	
		MCWB	79.1	79.3	79.7	78.7	77.4	76.2	75.6	75.4	74.8	75.2	76.8	78.5	
	5%	DB	84.5	85.5	85.4	84.4	82.2	80.8	80.3	79.8	80.5	81.1	83.4	83.9	
		MCWB	78.6	78.9	79.1	78.0	76.7	75.2	74.6	74.3	74.0	74.3	76.4	77.8	
	10%	DB	83.4	84.5	84.4	83.3	81.0	79.8	79.1	78.5	79.2	80.1	82.3	82.8	
		MCWB	77.9	78.3	78.6	77.4	75.7	74.6	74.1	73.0	73.1	73.5	75.9	77.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	81.0	81.9	81.3	79.7	78.9	78.2	77.9	77.9	78.3	79.9	81.3
			MCDB	85.3	85.5	86.1	85.3	83.0	82.0	81.5	81.1	81.2	81.7	83.3	84.5
2%		WB	80.1	80.3	80.8	79.9	78.4	77.7	77.0	76.8	76.3	76.8	78.8	80.0	
		MCDB	84.0	84.9	84.9	83.9	81.8	80.6	80.2	79.6	79.7	80.3	82.3	83.2	
5%		WB	79.3	79.6	80.0	79.0	77.7	76.7	76.0	75.6	75.3	75.8	77.9	79.1	
		MCDB	83.2	84.1	83.9	82.8	81.0	79.5	79.1	78.6	78.7	79.5	81.5	82.4	
10%		WB	78.6	78.9	79.2	78.2	76.9	75.8	75.1	74.5	74.5	74.8	77.1	78.3	
		MCDB	82.5	83.4	83.0	82.0	80.2	78.7	78.2	77.7	78.0	78.7	80.8	81.5	
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	7.1	6.7	6.2	6.1	5.9	6.5	7.0	6.9	7.2	6.9	
		MCDBR	7.6	7.4	7.8	7.6	6.9	6.5	6.7	7.5	7.8	7.7	8.0	7.6	
		MCWBR	4.3	3.9	4.3	4.3	4.1	4.0	4.3	4.5	4.2	4.3	4.1	4.0	
	5% WB	MCDBR	6.8	7.1	7.5	7.0	6.4	5.9	6.3	7.0	7.6	7.4	7.2	6.9	
		MCWBR	4.3	4.2	4.5	4.5	4.2	4.2	4.7	4.6	4.6	5.1	4.2	4.1	
Clear-Sky Solar Irradiance	taub		0.381	0.382	0.376	0.381	0.380	0.370	0.369	0.369	0.386	0.380	0.381	0.384	
	taud		2.639	2.635	2.660	2.640	2.623	2.629	2.623	2.615	2.564	2.589	2.606	2.612	
	Ebn at Noon		305	302	297	283	272	269	272	283	289	299	303	305	
	Edh at Noon		32	32	30	29	27	26	27	29	32	33	33	33	
All-Sky Solar Radiation	RadAvg		1959	1924	1800	1540	1310	1250	1307	1542	1798	1931	2030	1995	
	RadStd		168	141	146	112	114	70	67	88	98	163	148	168	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page